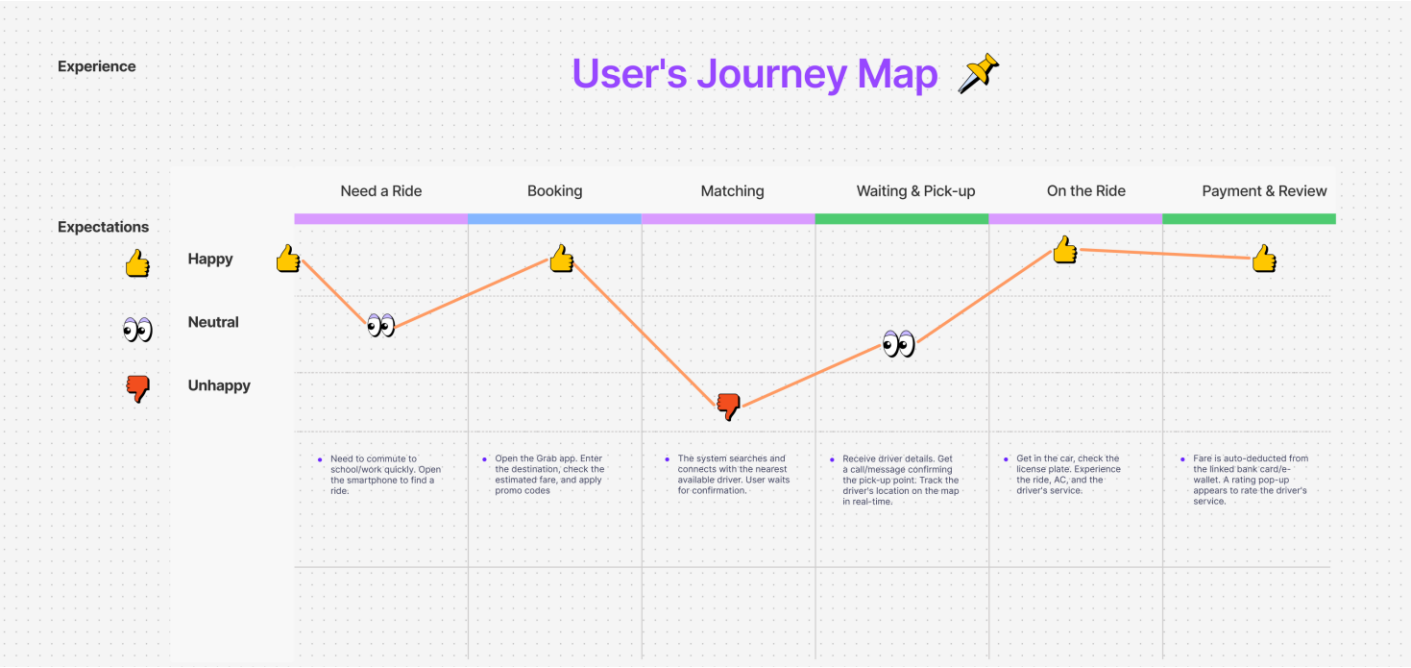
1. INTRODUCTION AND OBJECTIVES

In the digital era, user experience (UX) extends beyond a single screen; it is a continuous journey across multiple touchpoints, blending both online and offline environments. Grab is a prime example of a complex omnichannel ecosystem where user data flows seamlessly between the mobile app, backend servers, driver-partners, and notification channels. Analyzing Grab's cross-channel UX helps us understand how the system maintains continuity, handles edge cases, and ensures overall software quality and service excellence.

2. USER JOURNEY MAP



2.1. Pre-service Stage (Need a Ride and Booking) Every journey begins with a real-life mobility need, such as commuting to school or work during rush hour. At this stage, the user opens the mobile app. The UX focuses on minimizing clicks to fulfill a request. The user enters the destination; the system immediately calculates the distance, routes the optimal path using algorithms, and returns an estimated fare. The emotional state here is highly positive if the interface responds smoothly without latency when loading maps or applying promo codes.

2.2. During-service Stage (Matching, Waiting, and Riding) This is the most complex phase due to the strong intersection of digital and physical environments. First is the "Matching" state, where the backend system continuously loops to ping nearby drivers. If this takes too long, users may feel anxious. Once a driver is found, the touchpoint shifts to communication: users receive a Push Notification with the vehicle's license plate. In poor network conditions, the smart system falls back to SMS to ensure uninterrupted information flow. When the passenger boards the car, the experience transitions entirely to the physical environment (vehicle cleanliness, driver's attitude). The app then acts as a background live-tracking tool for safety.

2.3. Post-service Stage (Payment and Review) When the ride concludes, the payment process runs automatically via linked cards or e-wallets, providing a seamless checkout experience. Users do not need to pull out cash. Immediately, a rating pop-up appears on the screen. Simultaneously, an e-receipt is generated and sent to the user's email, completing a full experience loop.

3. LIST OF TOUCHPOINTS

To build the above journey, the system utilizes an interconnected network of touchpoints:

- **Mobile App (Core Application):** The primary digital touchpoint handling all client-side business logic, such as GPS tracking, booking, in-app calls/chats, and payment processing.
- **Physical Environment (Vehicle and Driver):** The touchpoint that dictates the actual service quality. This includes brand recognition (green uniforms, helmets) and driver professionalism.
- **Multichannel Notification System (Push/SMS/Email):** Acts as information checkpoints. Push Notifications report real-time status; SMS serves as a fallback during connection loss; Emails act as official financial records.
- **Call Center:** The touchpoint for crisis management and handling failed edge cases on the app, such as lost items or incorrect fare deductions.

4. CROSS-CHANNEL UX EVALUATION

4.1. UI Consistency Grab maintains an outstanding level of UI/UX consistency. Its Design System is strictly enforced. From the buttons and icons on the app to the layout of the email receipts, everything adheres to the primary green and white

color palette. This consistency even extends to the real world through the driver's uniform, reducing the user's cognitive delay when spotting their ride on a crowded street.

4.2. Continuity Across Devices User state data is continuously synchronized two-way. If a user messages a driver on the app, the driver receives and replies instantly. Continuity is also evident when a user logs into a new device; all ride history, credit card info, and saved promo codes are immediately restored, ensuring an uninterrupted flow.

4.3. Session Management (Resuming where left off) The system handles session management exceptionally well. In a common scenario: a user is typing a destination but accidentally minimizes the app to reply to a message. Upon returning to Grab, the system temporarily saves the previous state. Users do not have to start over. Or, if the app crashes mid-ride, reopening it immediately displays the active trip map instead of the default home screen.

5. STRENGTHS, LIMITATIONS, AND RECOMMENDATIONS

5.1. Strengths

- **Seamless and Automated Flow:** Successfully integrating maps, communication systems (with masked phone numbers for privacy), and payment gateways into a single platform. Users never have to leave the app.
- **Excellent Handling of Data Blind Spots:** When a driver enters a basement (losing GPS signal), the system uses interpolation or notifies

the user about the temporary signal loss, effectively managing user expectations.

5.2. Limitations

- **Cognitive Overload on the Home Screen:** Cramming too many secondary services (Food delivery, Mart, Insurance) into the main interface distracts users and increases the time needed to find the core "Transport" button.
- **Notification Spam:** The notification algorithm is not fully optimized. For the same promotional campaign, the system sometimes pushes alerts simultaneously via App, Email, and Zalo, causing unnecessary annoyance.

5.3. Recommendations for Improvement

- **Personalized Interface Customization:** Allow users to customize their home screen. If a user only commutes daily, they should be able to hide irrelevant services to make the app lighter and faster to load.
- **Notification Throttling Algorithm:** Build a logic flow to check the user's read status. For example, if the database logs that a user has clicked a promo notification on the app, the backend should automatically cancel identical Email or SMS triggers to prevent inbox spam.
- **Lock Screen Widgets (Live Activities):** Implement a feature to display live trip status (license plate, ETA) directly on the phone's lock screen. This allows users to track their ride without repeatedly

unlocking their phone, significantly optimizing usability during the waiting phase.